

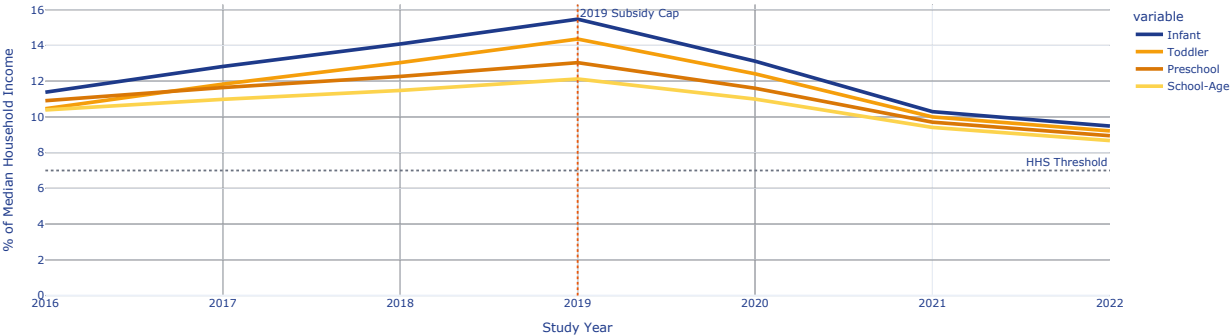
Papillion Families Can Reclaim Time & Money with Four-Day Weeks

Learn more at www.papillion-4-day.com



Is Childcare Affordable?	Subsidy Caps Helped, But Families Could Be Saving More	Time and Money Saved Simulator
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Papillion Families Still Pay 1.4x HHS Affordability Threshold Despite 2019 Subsidy Caps



Data: Sarpy County Childcare 2022 + 4-Day Week Global Trials | Papillion, NE



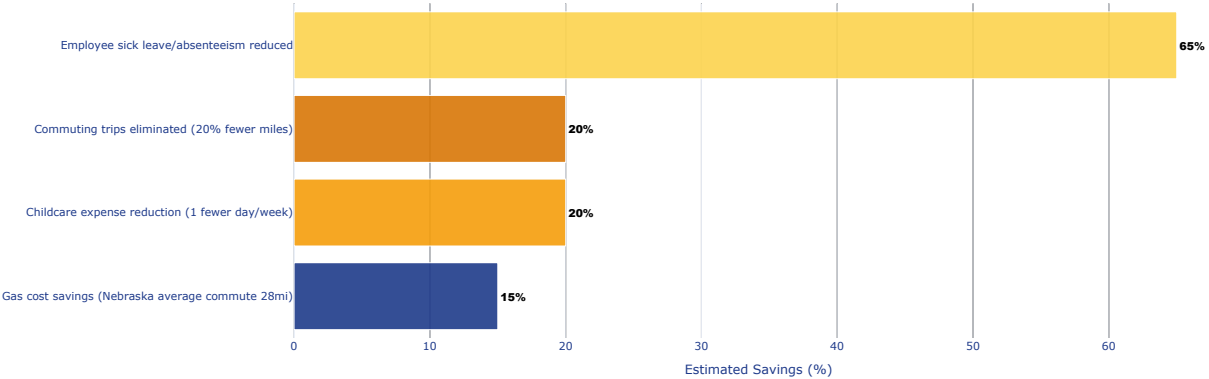
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Families Save with a Four-Day Week



Data: Sarpy County Childcare 2022 + 4-Day Week Global Trials | Papillion, NE

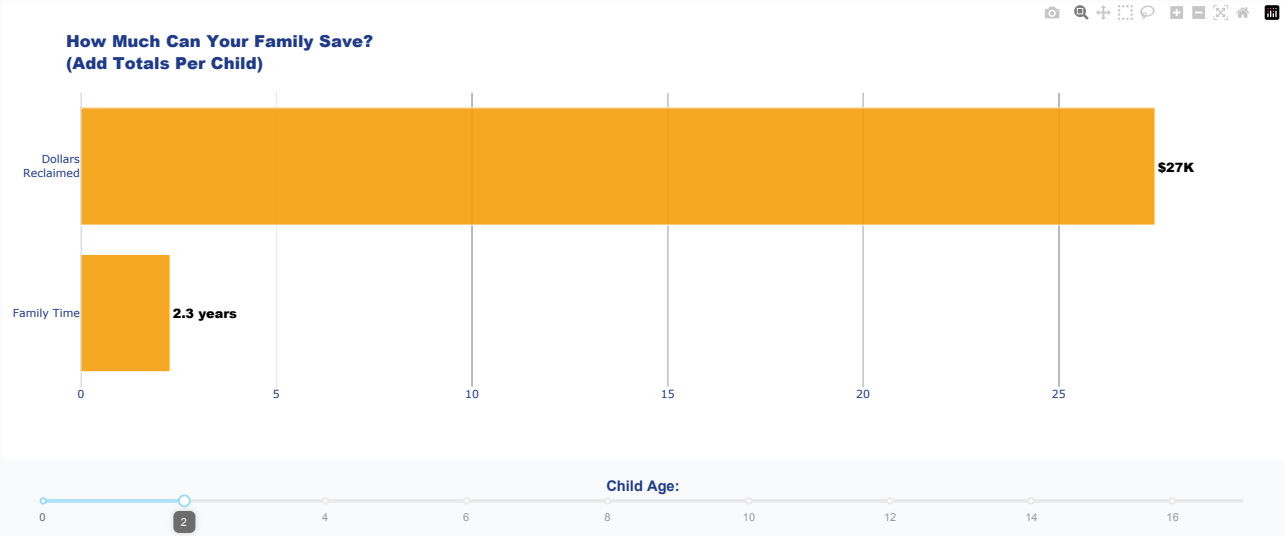


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- Is Childcare Affordable?
- Subsidy Caps Helped, But Families Could Be Saving More
- Time and Money Saved Simulator



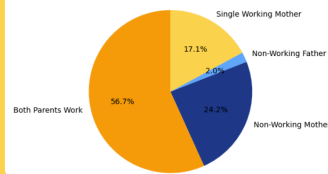
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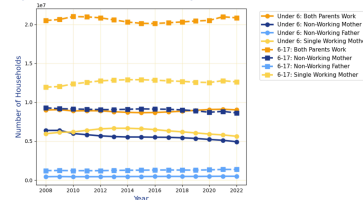
Papillion Parents Work...

Non-working parents are actually the minority.

Only 26.2% of Papillion Households with Children Under 6 Contained a Non-Working Parent in 2022



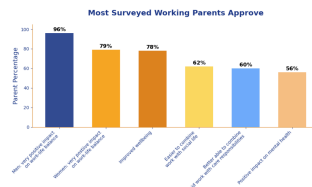
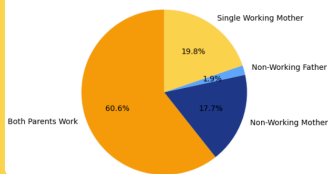
Papillion Household Trends Hold Steady from 2008-2022



This trend has held strong for over a decade.

Non-working parents are even less likely in households with children aged 6 and up.

Only 19.6% of Papillion Households with Children Ages 6-17 Contained a Non-Working Parent in 2022



Four-day work and school weeks support working parents.

Papillion should have a week that works for them.
Learn more at www.papillion-4-day.com.

Summary of Analysis & Findings

Exploratory data analysis revealed that the proportions of the four household types (both parents working, only the father working, only the mother working, and only single mothers working) varied only slightly between 2008 and 2018. The number of single-mother households grew slightly in both age groups (under 6 and 6-17), and the percentage of households with only fathers working dropped slightly in both age groups as well. Even with these minor changes in household composition, I was still struck by the fact that about 70% of households in the United States do not include a nonworking parent. That means approximately 70% of households rely on outsourcing their childcare, yet our public infrastructure is still designed for the traditional model of decades past, in which one parent is primarily responsible for the home and child-rearing.

As childcare costs continue to rise and the overwhelming majority of parents continue to work, the burden rate for the median income has surpassed the Department of Health and Human Services' definition of affordable childcare, which is 7% (Child care cost burden, n.d.). Anything above 10% is no longer considered affordable, which is alarming, given that the median cost burden of toddler and infant yearly care exceeded this threshold beginning in 2008. By performing rudimentary calculations, I found that reducing the standard five-day workweek to four days would reduce the childcare cost burden by 20%, allowing families to again achieve the DHS's definition of affordable childcare.

Assumptions & Items to Clarify

The price data includes imputations based on the best possible assumptions the U.S. Department of Labor – Women's Bureau could make from the available information. Despite the imputations, the data remain incomplete, requiring communication alongside any conclusions. For example, my project zoomed in on Sarpy County, which didn't have any data available before 2016, so all visualizations and calculations are representative only of 2016-2022, with the inclusion of their newest dataset publication.

Sarpy County has a higher median income than the national average, but there is still stark income disparity in the region. For this reason, the types of childcare facilities available to the average Sarpy County parent are varied. There are childcare centers that look like they're in desperate need of a building inspection, and are located next to scary, broken-down churches. There are childcare centers that charge more than your mortgage and still don't provide basic necessities for the children, such as diapers or meals (which means all these costs are in addition to the hefty tuition). These types of centers also tend to be closed for many days out of the year (about 30 days), during which the parent has to pay for care somewhere else, use their sick or vacation time, or take the hit to their paycheck and possibly progress at work. The lower-cost options in the county are not most Papillion parents' first choice, so the analysis uses the 75th percentile column for most calculations, as a balance between representing the facilities the higher-earning majority parents are most likely to choose and accounting for the hidden costs not included in the sticker tuition price within the datasets.

The four-day workweek calculation assumes median household incomes will remain unchanged, as has been the case in the majority of four-day workweek trials worldwide. They follow what's called a 100/80/100 model, which means employers deliver 100% of the pay for 80% of the hours but 100% of the productivity (4 Day Week research reports, n.d.). The calculation also assumes that the childcare centers will not increase their prices due to the four-day work week. Much of the cost burden in running a childcare center is wages, so if they are using the same 100/80/100 model, they would still need to pay caregivers 100% of their wages. But they would save money on utilities and other necessary supplies. They would likely need policy support and funding to sustain a transition to a four-day work week, unless they find another way to use the resources they already have to supplement the income.

Story Direction & Plan of Attack

After analysis and proposed solutions, the plan is to petition the public to urge the local school and state legislators to support a four-day workweek and school-week trial in Papillion, NE. The four-day workweek is already being implemented in many countries and in some U.S. corporations to improve work-life balance. According to the results of 4 Day Work Week Global's largest UK trial, 92% of companies opted to continue with the four-day work week, indicating that companies also see benefits (4 Day Week research reports, n.d.). Combining a four-day work week with a four-day school or childcare week could help address many issues, including the rising cost burden of childcare.

Target Audience

The target audience for the project is parents in Papillion, NE, and the surrounding areas included within the Papillion-La Vista Community Schools district. This means that all visualizations are being created for an audience that is likely unfamiliar with the data but has firsthand experience raising children in the area. As a member and fellow parent in the community, I've gained credibility through my experience as a Papillion parent. I understand how difficult it is to be a working parent, and I will be framing the visualizations to be relatable with this understanding.

Mediums

I chose to include three different media for the project, each with a call-to-action link to "Learn More at www.papillion-4-day.com". The first medium is an interactive dashboard. This medium is meant to be shared on a cell phone or laptop. It helps users gain a deeper understanding of the problem and the solution, as well as the direct impacts each can have on their own lives. The second medium is an infographic that can be distributed via print or digitally for versatility and maximum exposure. It focuses on how Sarpy County household types rarely contain a non-working parent, and how that trend has held

steady in recent decades. The medium proposes a solution that works for working parents. The third medium is a video to be shared via social media, or in other spaces where it would make sense, such as a PTA meeting. The video tells a story about how working parents report higher levels of stress, and how it feels like there's never enough time in the day. It then proposes a solution that allows parents to reclaim financial resources and time while reducing stress through a four-day work/school week. Short videos can have a lasting impact on viewers and can be easily circulated in today's social media age.

Design Decisions

Each medium was created using the same color palette, drawn from the Papillion city logo. I wanted the media to feel like they were part of the same campaign, so that if you saw one medium after another, you would know they came from the same place. I did this because I wanted to message to feel cohesive, while still feeling familiar to a Papillion resident who often sees the colors and logo around them in the community. I've also added the Papillion logo itself to two of the mediums, as the final image in the movie and alongside the call to action in the dashboard to boost familiarity. The colors are different shades of blue, orange, and yellow, which means the media are designed to be bright and inviting. Every chart is accompanied by an explanation to guide the viewer through the story it's telling, so they can make their own choice on how much time they'd like to spend interpreting the charts.

Ethical Implications

I chose to focus on only a subset of the data, so much of the information from the dataset as a whole isn't communicated in this project. As previously mentioned, the dataset included missing values and imputations, so even the assumptions about the available dates aren't perfect. The dataset itself was ethically sourced, but there are nuances that aren't captured in the values. Choosing to use the 75th percentile for the calculations is risky, but it is a decision I made based on my own experience as a Papillion

parent and on conversations with fellow Papillion parents. That said, each Papillion parent knows how much they're paying for care, and they will factor that cost into their overall conclusions about the media. The data included for the pilot programs was collected and compiled using Perplexity AI, and while the figures are accurate based on the source pages, a more comprehensive compilation of all pilot programs to date and their outcomes would have led to a more accurate analysis.

Lessons Learned

Overall, I really enjoyed the creative aspect of this project. This is the first time I've been able to create multiple media that feel cohesive in this program. While there were many bumps along the road, seeing the end result of where Data Science can take you visually is very satisfying. If I were to do it again, I would incorporate modeling into my analysis, using more extensive, detailed data from the pilot projects to predict Papillion-specific results for doppelganger communities.

References

4 Day Week research reports. 4 Day Week Global. (n.d.). <https://www.4dayweek.com/research>

Child care cost burden. County Health Rankings & Roadmaps. (n.d.-b).

<https://www.countyhealthrankings.org/health-data/community-conditions/social-and-economic-factors/safety-and-social-support/child-care-cost-burden?year=2025>